

Name: _____ **Date:** _____

Problem 1. A spherical balloon is inflated and its volume increases at a rate of 15 inches per minute. What is the rate of change of its radius at the point in time when the radius equals 10 inches? (hint: a sphere of radius r has volume $V = \frac{4}{3}\pi r^3$)

Problem 2. A 13-foot ladder is leaning against a vertical wall when the foot of the ladder begins sliding away from the wall at a rate of 0.5 feet per second. How fast is the top of the ladder sliding down the wall at the point in time when the foot of the ladder is 5 feet from the wall? (hint: draw a picture!)

Problem 3. A child is standing motionless on the ground, holding a ball of string, attached to a kite flying 40 feet above the ground. The kite is flying away from the child, horizontally, at a rate of 10 feet per second. At what rate must the child release the string at the point in time when the string is 50 feet long? (hint: draw a picture!)

Problem 4. Suppose a building has an elevator that is visible from outside, and there is somebody standing 500 feet away from the building. Drawing an invisible line from where the person is standing to the elevator car creates an angle with the ground, which we call the *angle of elevation*. If the elevator is descending at a rate of 26 feet per second, at what rate is the angle of elevation changing at the point in time when the elevator is 700 feet off the ground?